

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
 School of Computer Engineering & Mathematical Sciences
 M.Tech End Semester- Spring 2023 Examinations

Roll No:

Date: 29-05-2023

Total Number of Questions: [05]

Course Title: Data Science: Tools and Techniques

Course Code: AM624D

Time Allowed: 3.0 Hours

Max Marks: [50]

Instructions / NOTE

- i. Attempt All Questions.
- ii. Support your answer with neat freehand sketches/diagrams, wherever appropriate.
- iii. Assume any missing data to suit the case / derivation / answer.

Q1 [10 Marks]	Explain the following terms: (a) Feedforward artificial neural networks (b) Difference between iloc and loc in the context of pandas dataframe. (c) Given $128 \times 128 \times 3$ and 11 filters of size $5 \times 5 \times 3$, what will be the dimension of the output volume when the stride=1 and no padding. (d) What is the main benefit of stacking multiple layers of neurons with non-linear activation functions over a single-layer perceptron? (e) Suppose a fully-connected neural network has a single hidden layer with 10 nodes. The input is represented by a 5D feature vector and the number of classes is 3. Calculate the number of parameters of the network. Consider there are NO bias nodes in the network.	[02] [02] [02] [02] [02]								
Q2 [10 Marks]	(a) Learn $y = \theta_0 + \theta_1 x$ on the following dataset, using gradient descent where initially $(\theta_0, \theta_1) = (4, 0)$ and step-size, $\alpha=0.1$ for 2 iterations. <table><tr><td>x</td><td>y</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table> (b) Propose a neural network model to realize a XNOR gate and compute the weights of the network to satisfy the truth table.	x	y	1	1	2	2	3	3	[05] [05]
x	y									
1	1									
2	2									
3	3									
Q3 [10 Marks]	(a) Using the log odds ratio, derive the Multi-class logistic regression cost function. (b) Explain the following data types with examples i) Nominal Data ii) Ordinal Data iii) Interval Data iv) Ratio Data?	[05] [05]								

Q4
[10 Marks]

(a) Write a python program that takes a number X and prints the multiplication of the individual digits of the input number.

Eg if X=123 the output is 6,
if X=1234 then output is 24

(b) Consider an image I (4 x 4 x 1) in a CNN given below:

10	0	3	2
0	10	5	4
3	5	10	5
2	4	5	10

Find the convolution with a filter K considering stride =1 and zero padding of width 1 pixel:

-1	0	1
0	-1	0
1	0	-1

Q5
[10 Marks]

(a) Consider the following weather model predictions and ground truth for a classification task. Compute the confusion matrix, accuracy, precision, recall for the dataset:

Predictions ($\hat{y}$)	Ground Truth (y)
Rainy N	Rainy TN
Rainy N	Dry P
Dry P	Rainy N
Dry	Dry
Dry	Dry
Dry P	Rainy N
Rainy	Rainy
Rainy	Rainy
Rainy	Rainy
Rainy N	Dry
Dry	Dry
Dry	Dry

(b) Two layer neural network, as defined below:

$$Z^{[1]} = W^{[1]}A_{(l)}^{[0]} + B^{[1]}$$

$$A^{[1]} = \text{ReLU}(Z^{[1]})$$

$$Z^{[2]} = W^{[2]}A^{[1]} + B^{[2]}$$

$$\hat{y}_{(l)} = \sigma(z^{[2]})$$

$$L_{(i)} = y_{(i)} * \log(\hat{y}_{(i)}) + (1 - y_{(i)}) * \log(1 - \hat{y}_{(i)})$$

$$J = \frac{-1}{m} \sum_{i=1}^m L_{(i)}$$

Note that $A_{(i)}^{[0]}$ represents single input example and is of shape $\mathbb{R}^{N^{[0]} \times 1}$ and $N^{[0]}$ is the number of input nodes in zeroth layer. Further $y_{(i)}$ is a single output label and is a scalar. There are m examples in our data set. We will use $N^{[1]}$ nodes in out hidden layer i.e. $Z^{[1]}$ shape is $\mathbb{R}^{N^{[1]} \times 1}$.

- i. What are the shapes of $W^{[1]}, B^{[1]}, W^{[2]}, B^{[2]}, X$ and Y assuming that we are vectorizing this network across multiple examples m ?
- ii. What is $\frac{\partial J}{\partial \hat{y}_{(i)}}, \frac{\partial \hat{y}_{(i)}}{\partial z^{[2]}}, \frac{\partial z^{[2]}}{\partial A^{[1]}}, \frac{\partial A^{[1]}}{\partial z^{[1]}}, \frac{\partial z^{[1]}}{\partial W^{[1]}}$?
- iii. What is $\frac{\partial J}{\partial W^{[1]}}$?